2.1 Offline Knob Programming

This section demonstrates how to program offline using a knob. That idea is that robot arm grasps a small wood block on the left, then rotate to the right to drop it. Finally, the pan-tilt servo returns to neutral.

1. Modes and Buttons Introduction

For the buttons function and mode instruction, you can refer to the following table:

Modes	Buttons	Effect
①Neutral mode (white light on)	The startup mode is the default mode. You can long press K1 and K2 at the same time to exit.	Servo ID 1~5 are in 90° attitude
②Normal mode (green light on)	Long press red button K1	Enter ③ and clear the original action group
	Short press yellow button K2	Enter ④ and perform preserved action group once
	Long press yellow button K2	Enter ④ and perform preserved action group in a loop
③Action group editing mode (red light on)	Short press red button K1	Preserve action
	Long press red button K1	Preserve and exit edited action group to switch to the

		normal mode
④Action group running mode (yellow light on)	Short press red button K1	Stop running action group and switch to the normal mode

2. Knob Programming

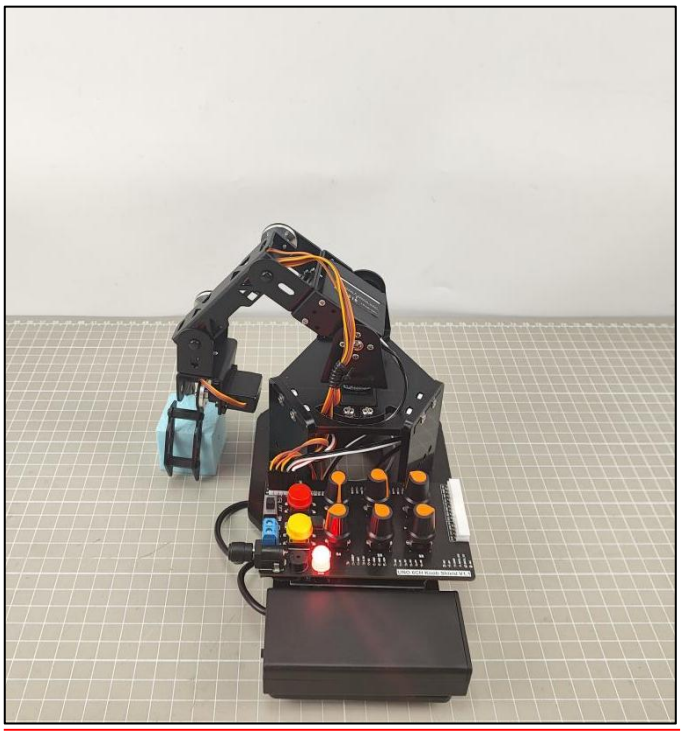
1)Firstly, open robotic arm. Long press buttons K1 and K2 on the expansion board and release them till the green light is emitted. Then wait for 2 seconds for servo to return to the knob control attitude.

2)When you long press the red button K1 on the expansion board, the buzzer will emit a prompt sound and the RGB light will change from green to red. At this point, the robotic arm will clear the the action group

preserved in the control board and enter the action group editing mode.

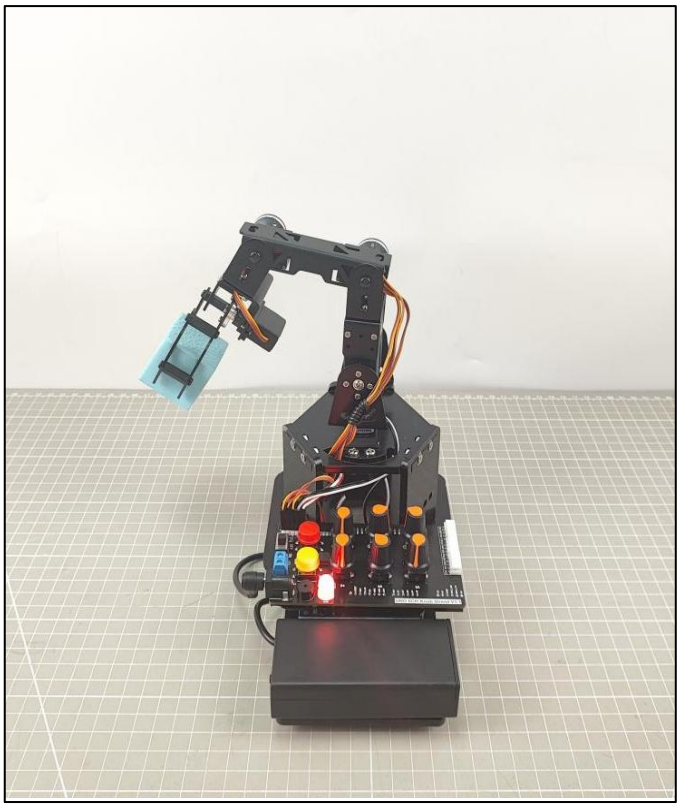
3)Rotate knobs S1-S5 to turn the robotic arm to the left, so that the robotic arm is in the grasping state. Then short press K1, the buzzer emits a short sound to preserve the first action.

4)Next, place the small block under the robotic claw. And rotate knob S1 to close the robotic claw to grab the small block. Then short press red button K1, the buzzer emits a short sound to preserve the second motion.



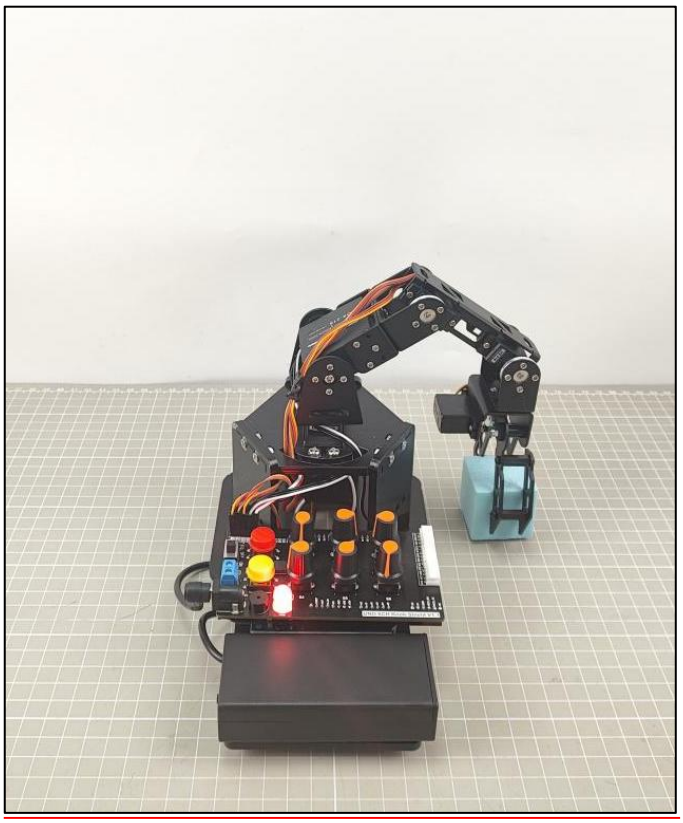
5)Rotate knob S4 to lift robotic arm. Then short press red button K1, the

buzzer emits a short sound to preserve the third motion.

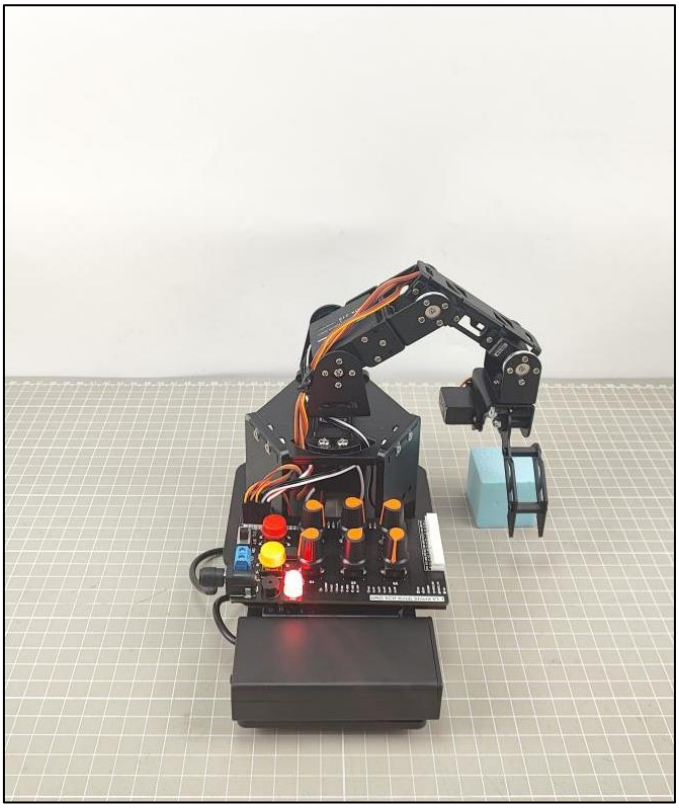


6)Rotate knob S5 to turn the robotic arm to the right side. Then short press red button K1, the buzzer emits a short sound to preserve the fourth motion.

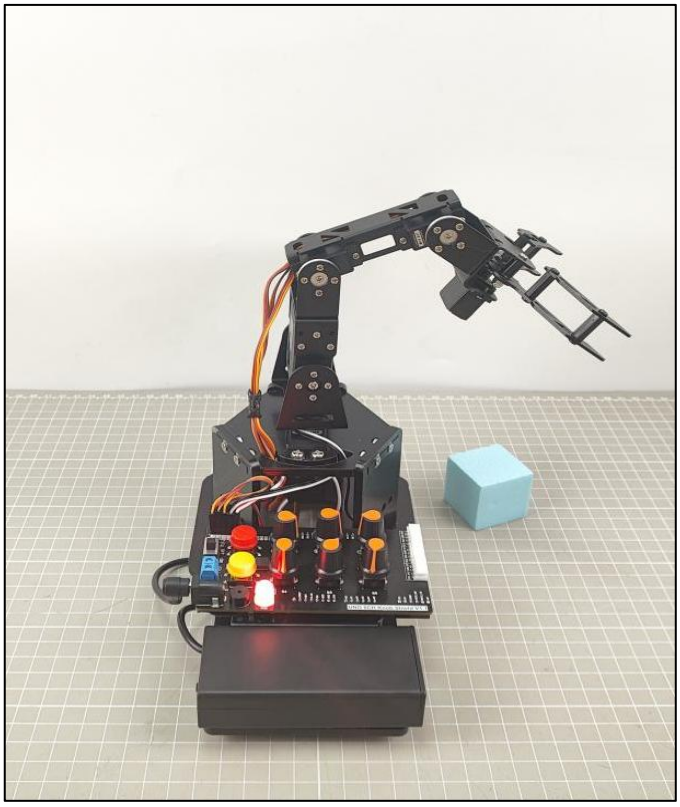
7)Rotate knob S4 to control the lowering of the robotic arm. Then short press red button K1, the buzzer emits a short sound to preserve the fifth motion.



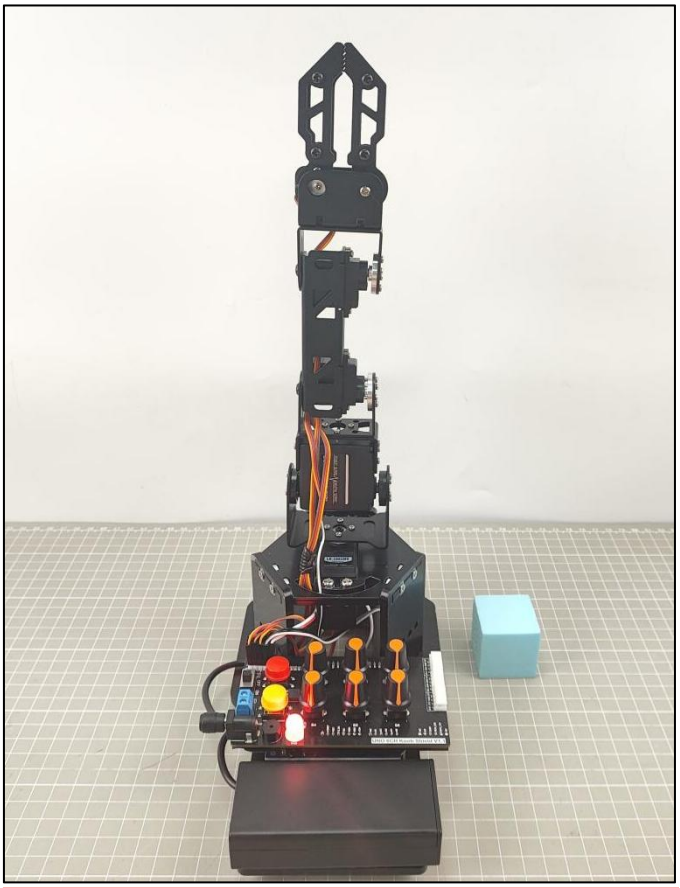
8)Rotate knob S1 to release small block. Then short press red button K1, the buzzer emits a short sound to preserve the sixth motion.



9)Rotate knob S4 to lift the robotic arm. Then short press red button K1, the buzzer emits a short sound to preserve seventh motion.



10)Rotate knob S1-S5 to return the robotic arm to the initial position. Then short press red button K1, the buzzer emits a short sound to preserve the eighth motion.



11)Finally, long press red button K1, the buzzer emits a long sound and RGB will switch to green. At this point, the robotic arm will preserve current edited action group and exit this mode.

Note: long press the red button K1 to enter the action group editing mode. This action will also simultaneously clear previously edited action group data.

3. Run Action Group

After editing action group, you can perform preserved action group through pressing yellow button K2 on the expansion board. Long-pressing can run the action group in a loop.

Note:

1. The edited action group can be preserved after power-off, which means that they can be executed again upon the next power-on.

2. Short press red button K1 to switch to the normal mode if you want to stop the running action group.